The Transported Probability Density Function Method and its Relation to the Black-Scholes-Merton Equation

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Financial Derivatives: Options

Derivatives: A financial instrument whose value depends on (or derives from) the values of other, more basic, underlying variables.

Stock Options: A derivative whose value is dependent on the price of a stock.

Call Option: Gives the holder the right, but not the obligation, to buy the underlying asset by a certain date for a certain price.

Put Option: Gives the holder the right, but not the obligation, to sell the underlying asset by a certain date for a certain price.

European Options: Can only be exercised on the expiration date.

American Options: Can be exercised at any time up to the expiration date.

Strike Price: The price at which the asset may be bought or sold in an option contract.

Figure 10.3 Profit from writing a European call option on one share of a stock. Option price = \$5; strike price = \$100.

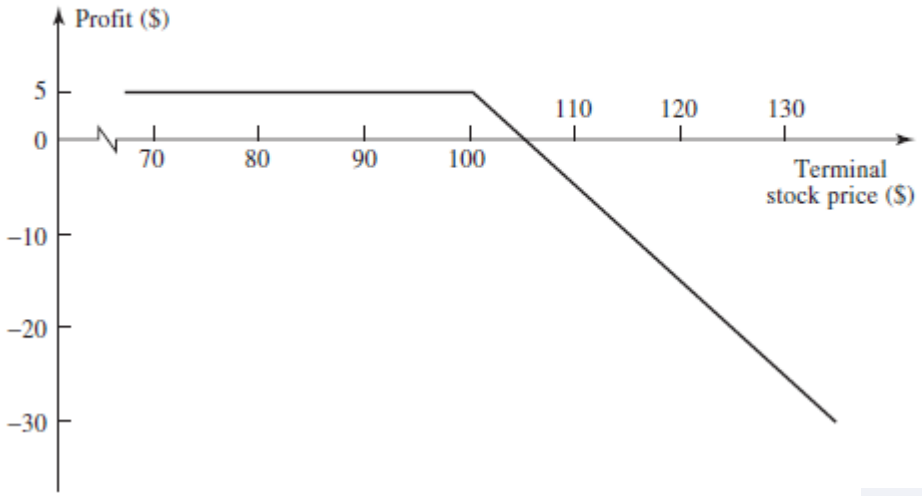
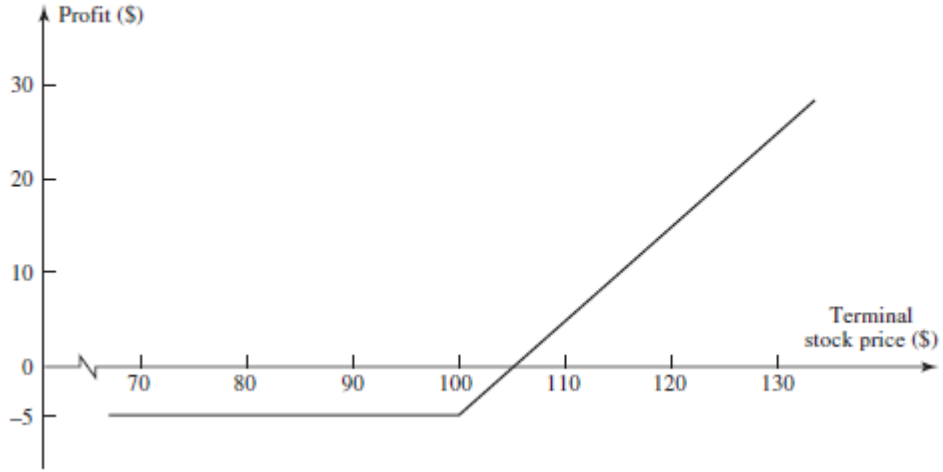


Figure 10.1 Profit from buying a European call option on one share of a stock. Option price = \$5; strike price = \$100.



$$V = \max(S_T - K, 0)$$

The Black-Scholes-Merton Equation

Black, Scholes (1973) and Merton (1973) independently derived the following PDE for pricing European options:

Black-Scholes-Merton PDE

$$\frac{\partial V}{\partial t} + \frac{1}{2}\sigma^2 S^2 \frac{\partial^2 V}{\partial S^2} + rS\frac{\partial V}{\partial S} - rV = 0$$

This equation can be derived from the Geometric Brownian Motion SDE for stock prices:

$$\mathrm{dS} = \mathrm{rSdt} + \sigma\mathrm{SdW}$$

V = option value *S = stock price* *r = risk-free rate* *σ = volatility* *dW = Wiener process*

Itô's Lemma:

$$\mathrm{dV} = \left(rS\frac{\partial V}{\partial S} + \frac{\partial V}{\partial t} + \frac{1}{2}\sigma^2 S^2 \frac{\partial^2 V}{\partial S^2}\right)\mathrm{dt} + \frac{\partial \mathrm{V}}{\partial \mathrm{S}}\sigma\mathrm{SdW}$$

The Transported PDF Method

In a turbulent flow, the velocity field $\boldsymbol{U}(\boldsymbol{x}, \mathrm{t})$ is random (Pope, 2000). The exact transport equation for the Eulerian PDF of the velocity, $\mathrm{f}(\boldsymbol{V};\ \mathrm{x}, \mathrm{t})$, derived from the Navier-Stokes equations is expressed as

$$\frac{\partial f}{\partial t} + V_i + \frac{\partial f}{\partial x_i} = -\frac{\partial}{\partial V_i}\left[f\ \left\langle \frac{DU_i}{Dt} \middle| \boldsymbol{V} \right\rangle\right]$$

The equation for the evolution of the conditional PDF of the particle velocity $\mathrm{f}^*(\boldsymbol{V}|\boldsymbol{x}; \mathrm{t})$ is expressed as:

$$\frac{\partial \mathrm{f}^*}{\partial \mathrm{t}} + \mathrm{V_i}\frac{\partial \mathrm{f}^*}{\partial \mathrm{x_i}} = -\frac{\partial}{\partial \mathrm{V_i}}\left[\mathrm{f}^* \mathrm{a_i}(\boldsymbol{V}, \boldsymbol{x}, \mathrm{t})\right] + \frac{1}{2}\mathrm{b}(\boldsymbol{x}, \mathrm{t})^2 \frac{\partial^2 \mathrm{f}^*}{\partial \mathrm{V_i} \partial \mathrm{V_i}}$$

Which can be derived from the following stochastic differential equation for a generalized Langevin Model using **Itô Calculus**

$$\mathrm{d}\boldsymbol{U}^*(\mathrm{t}) = \boldsymbol{a}(\boldsymbol{U}^*(\mathrm{t}), \boldsymbol{X}^*(\mathrm{t}), \mathrm{t})\mathrm{dt} + \mathrm{b}(\boldsymbol{X}^*(\mathrm{t}), \mathrm{t})\mathrm{d}\boldsymbol{W}$$

*a(**U***(t), **X***(t), t) = drift coefficient* *b(**X***(t), t) = diffusion coefficient* *d**W** = Vector Valued Wiener process*

The Connection

Both equations are derived from SDEs of the same drift + diffusion form.

Stock Price SDE

$$dS = rSdt + \sigma SdW$$

Langevin SDE

$$d\boldsymbol{U}^*(t) = \boldsymbol{a}(\boldsymbol{U}^*(t), \boldsymbol{X}^*(t), t)dt + b(\boldsymbol{X}^*(t), t)d\boldsymbol{W}$$

Variable Mapping

Finance (GBM / BSM)	CFD (Langevin / Fokker-Planck)
Stock price, S	Particle velocity U*
Risk-free rate, r	Drift coefficient a
Volatility, σ	Diffusion coefficient b
Wiener process, dW	Vector-valued Wiener process, d**W**
BSM PDE	Fokker-Planck (forward Kolmogorov)
Monte Carlo simulation	Lagrangian particle method

Solution Methods

Eulerian (Grid-Based)

Solve the PDE directly on a fixed grid

Methods: Finite Volume, Finite Difference, Finite element

Advantage: No statistical noise

Limitation: Scales poorly to high dimensions

Lagrangian (Particle-Based)

Simulate random trajectories

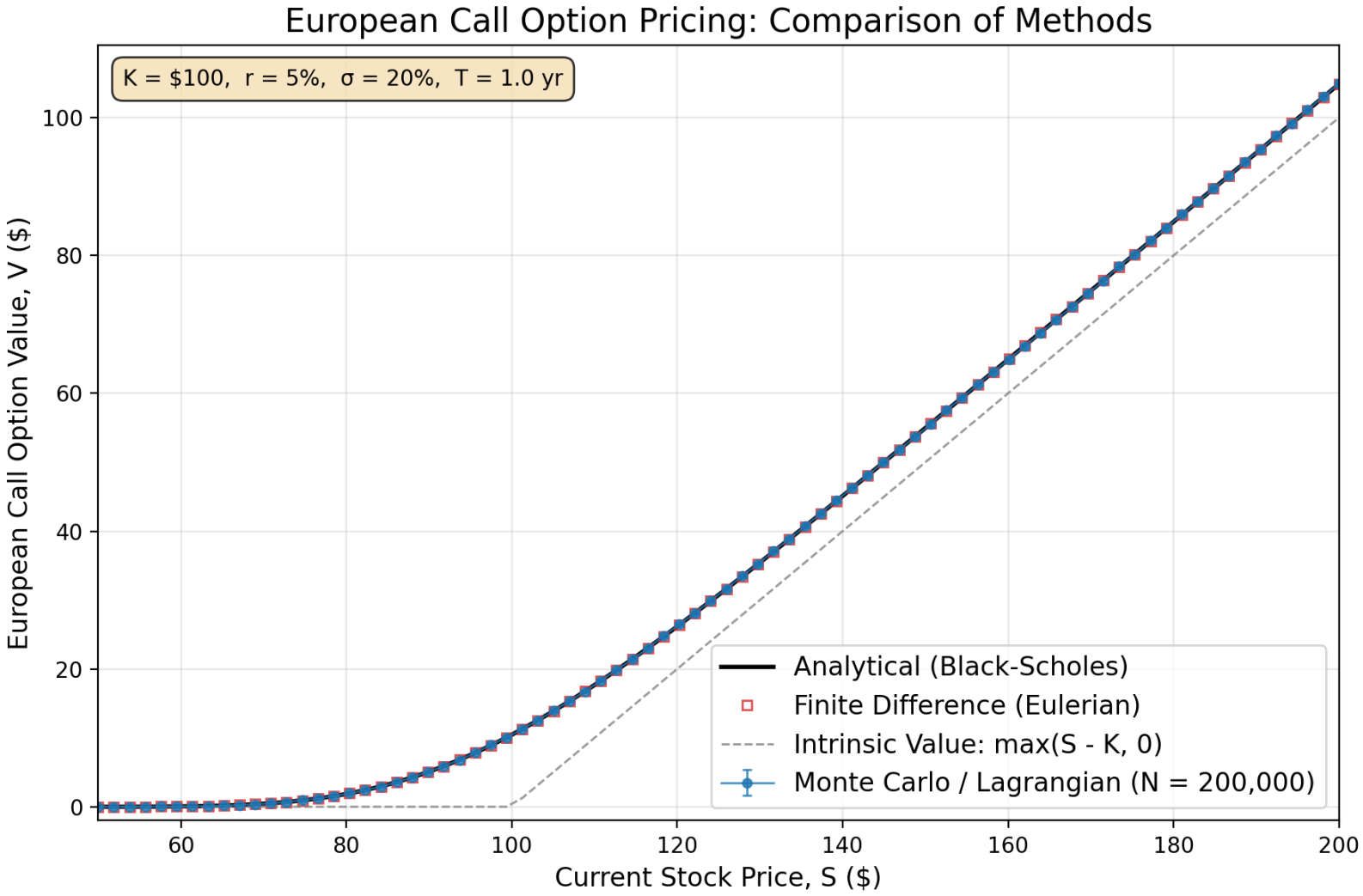
BSM: Monte Carlo simulation

CFD: Lagrangian particle method

Advantage: Scales to high dimensions, compatible with CPUs/accelerators

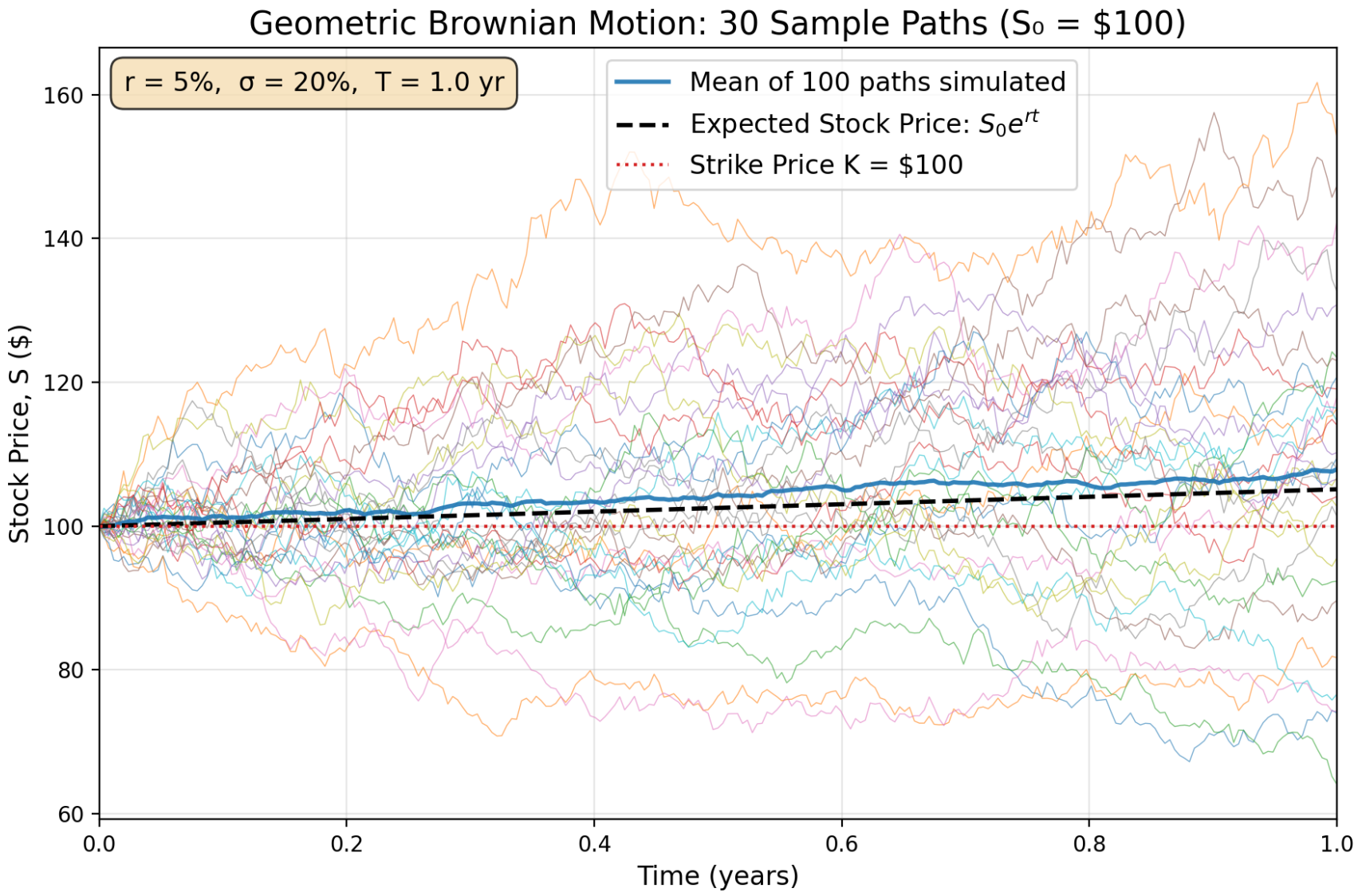
Limitation: Statistical noise, slow convergence $\left(error \sim 1/\sqrt{\mathrm{N}}\right)$, high computational cost

Results: Option Pricing

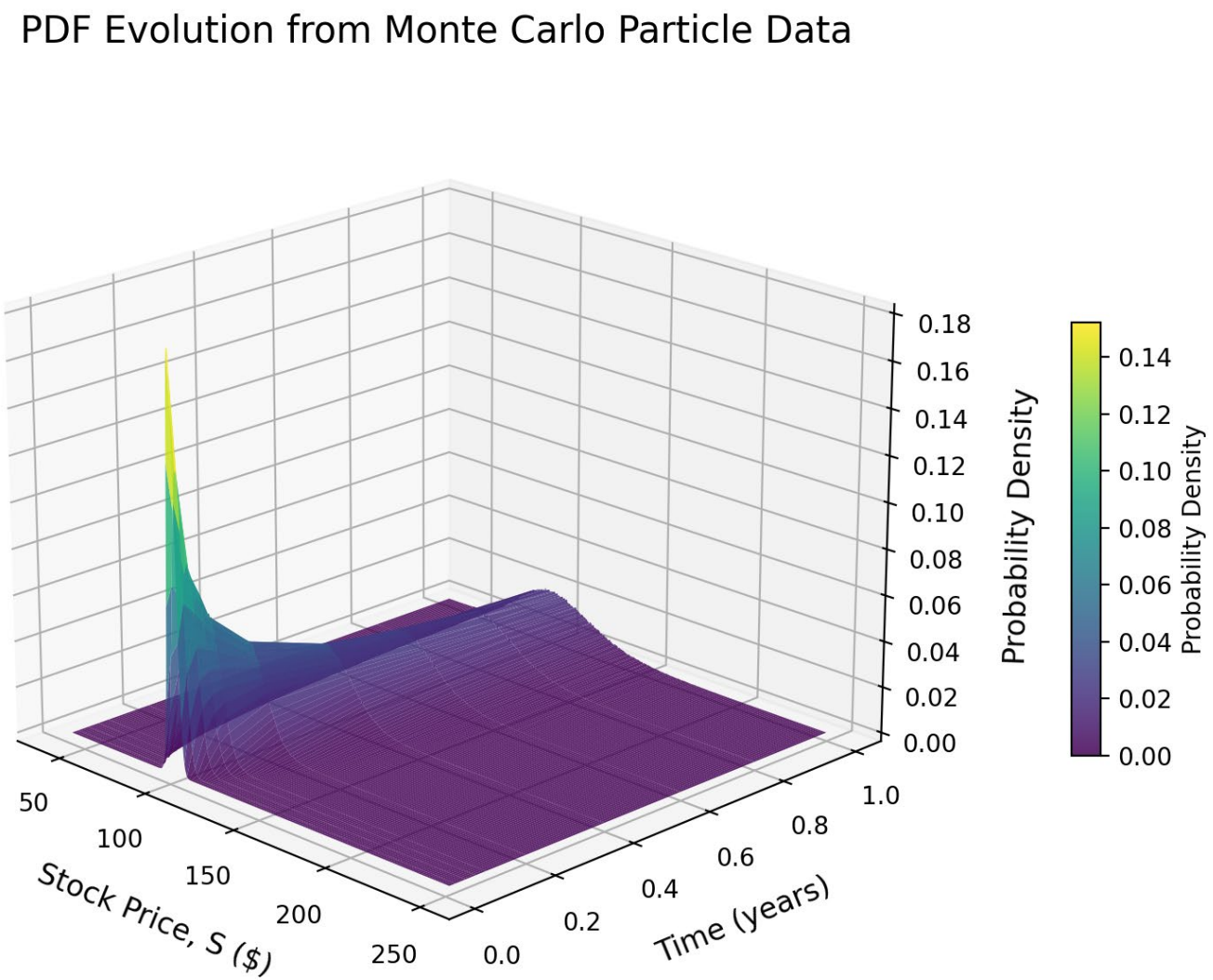


Analytical (Black-Scholes), Finite Difference (Eulerian), and Monte Carlo (Lagrangian) solution methods produce identical option values.

Results: Monte Carlo Visualization



Results: PDF Transport



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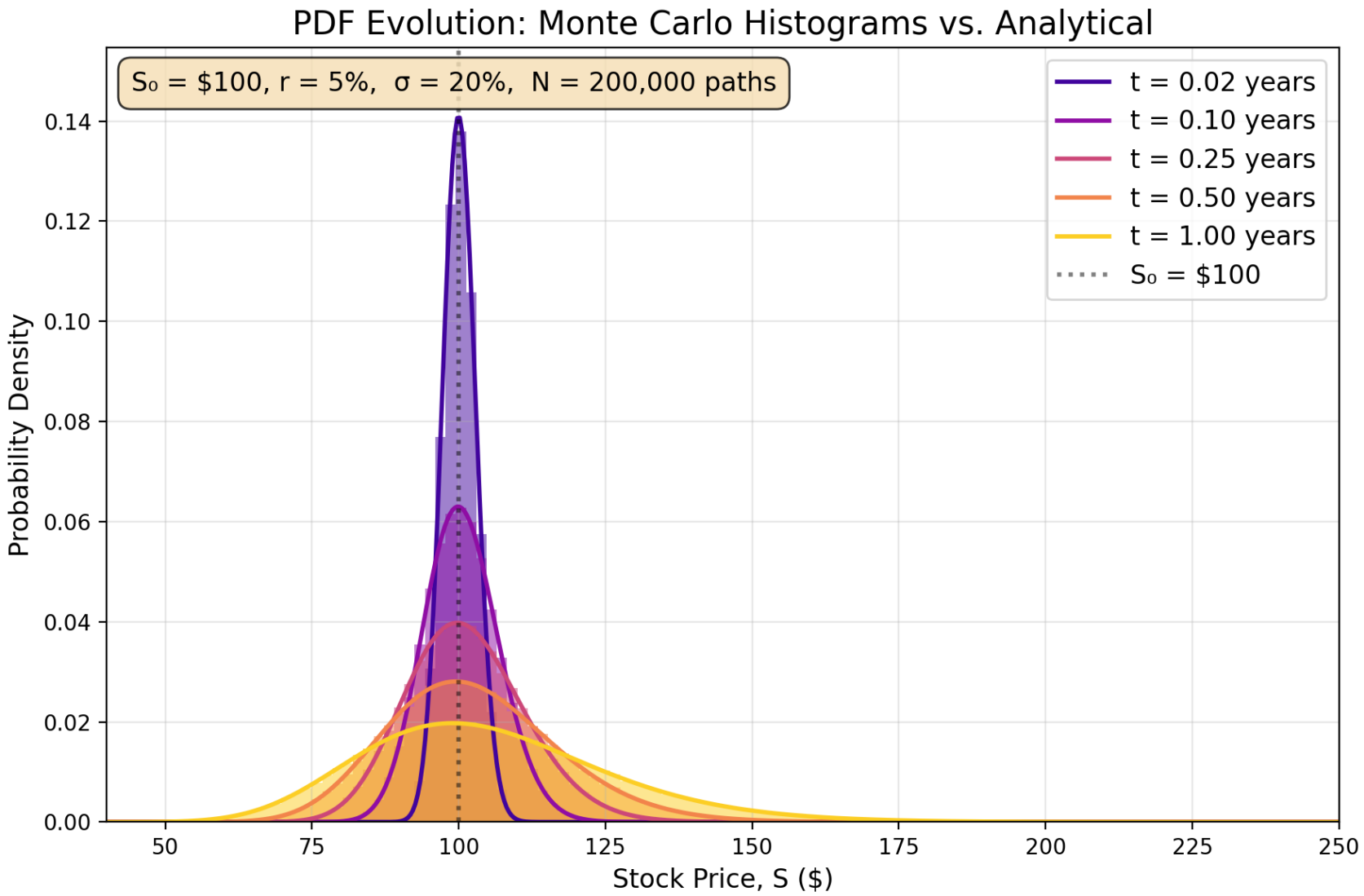

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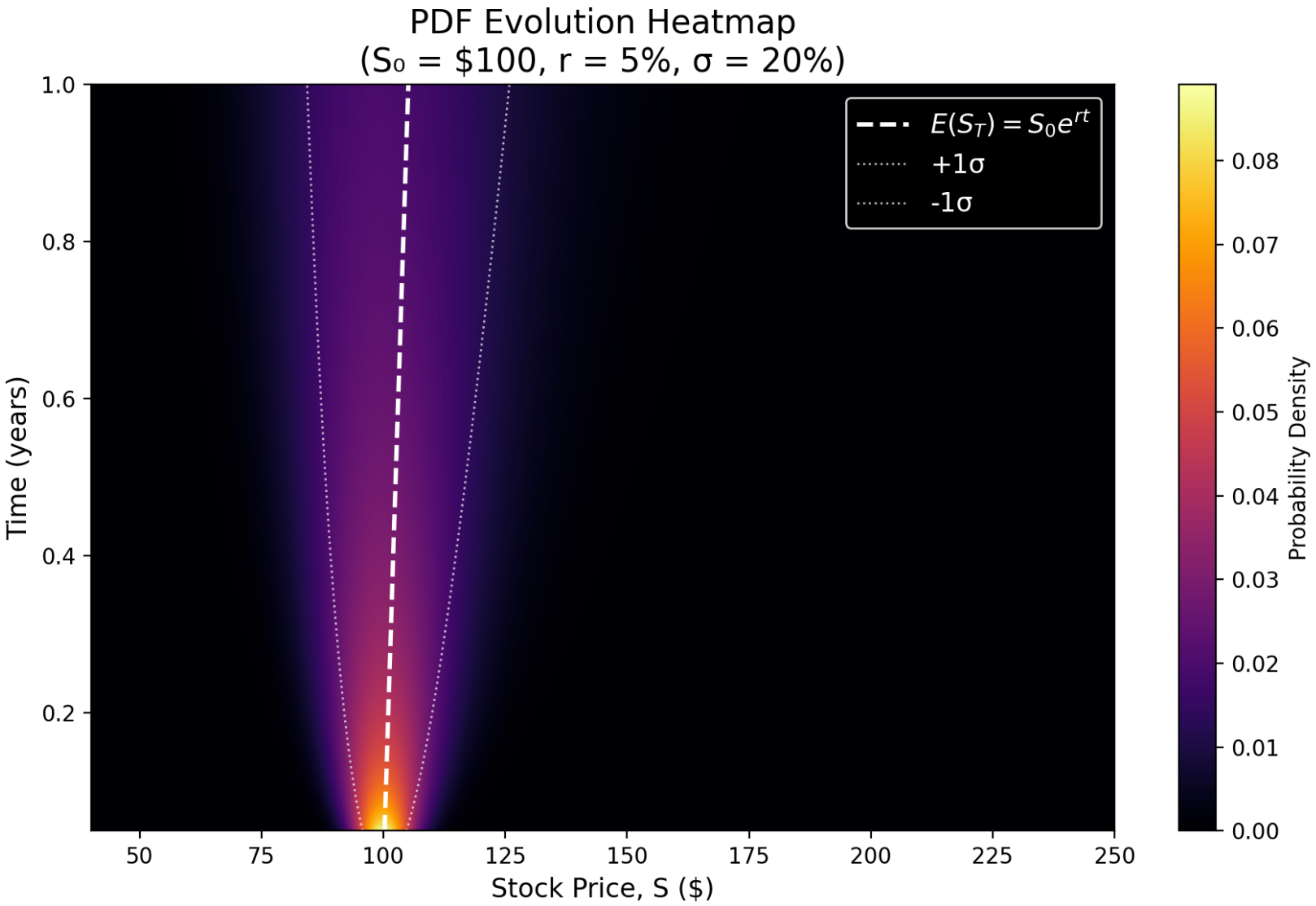
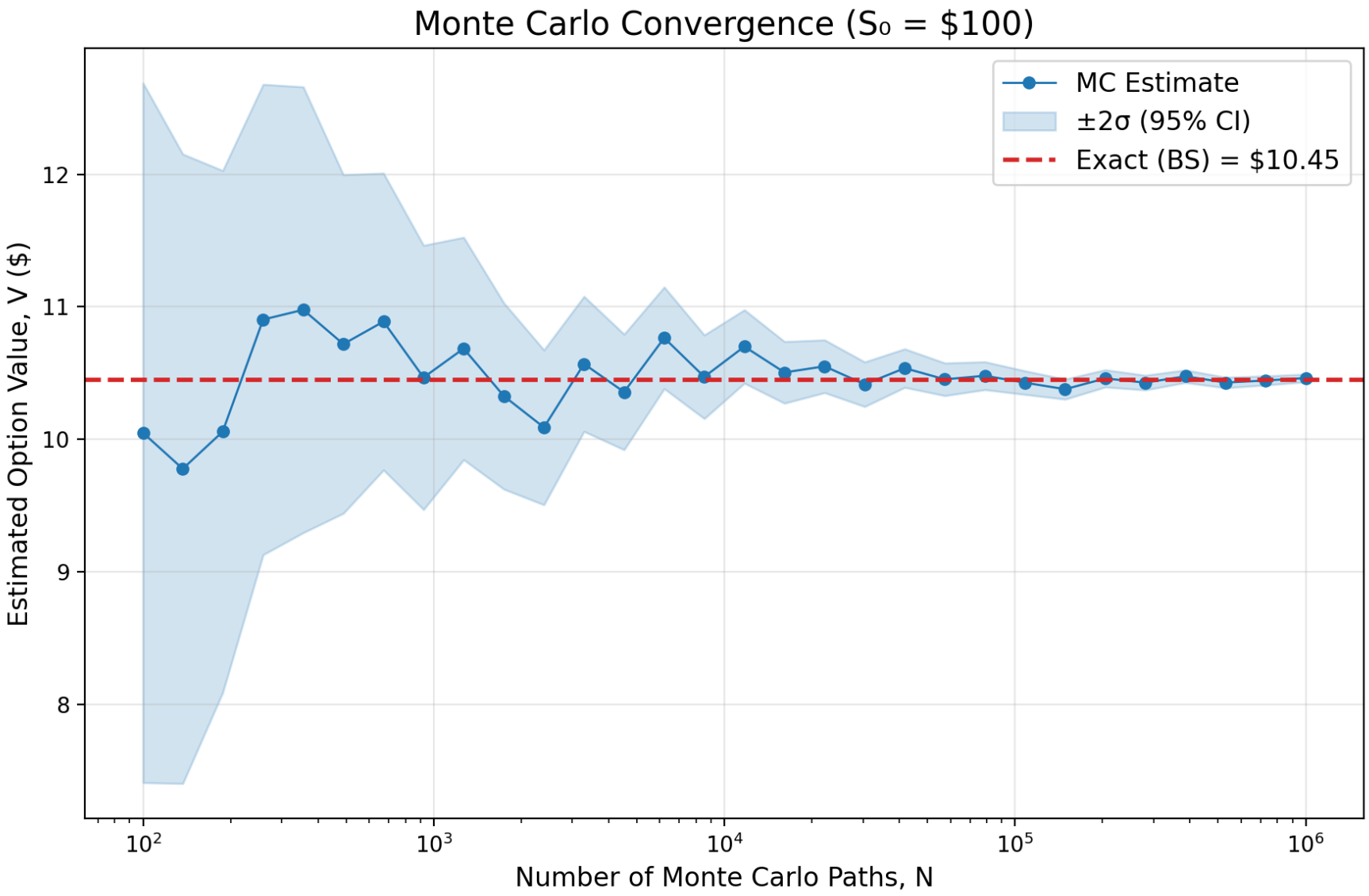


Comments: Monte Carlo Convergence



Conclusions

1. The Black-Scholes-Merton PDE and the Fokker-Planck equation (forward Kolmogorov) are both derived from SDEs of the same drift-diffusion form, establishing a direct mathematical connection between financial derivative pricing and the transported PDF method in CFD.

2. Both Eulerian (finite difference/finite volume) and Lagrangian (Monte Carlo/particle) methods produce results consistent with each other and with analytical solutions.

3. Monte Carlo particle simulations can reproduce the results consistent with analytical probability density functions.

4. The Black-Scholes-Merton equation can serve as a validation problem for Lagrangian particle methods because closed-form solutions exist for both the SDE and the PDF.

5. Advancements in financial modeling and computational fluid dynamics are mutually beneficial.

References

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